User Guide

for

Formula Subset Analysis (FSA)

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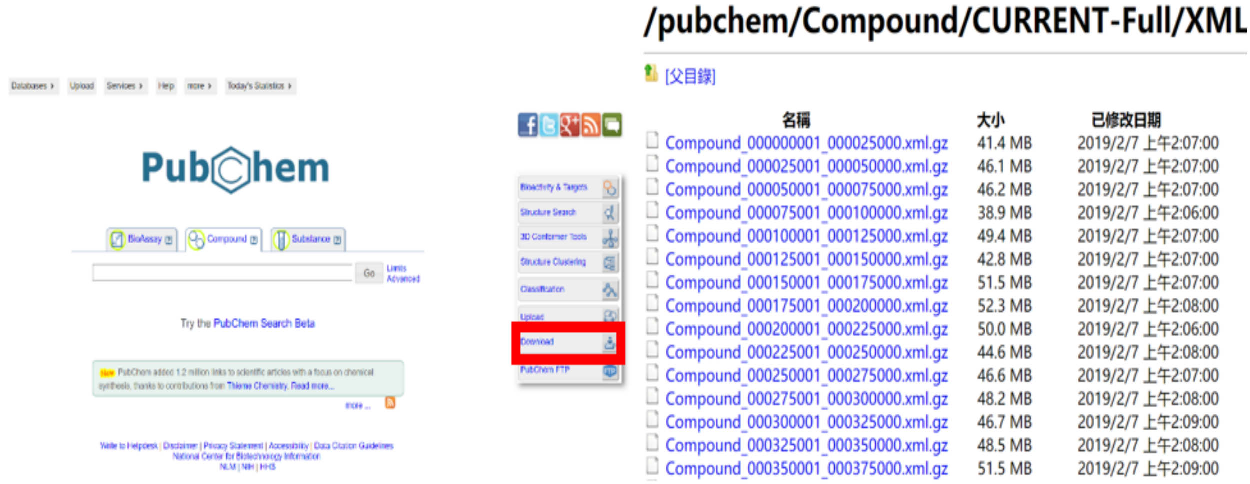
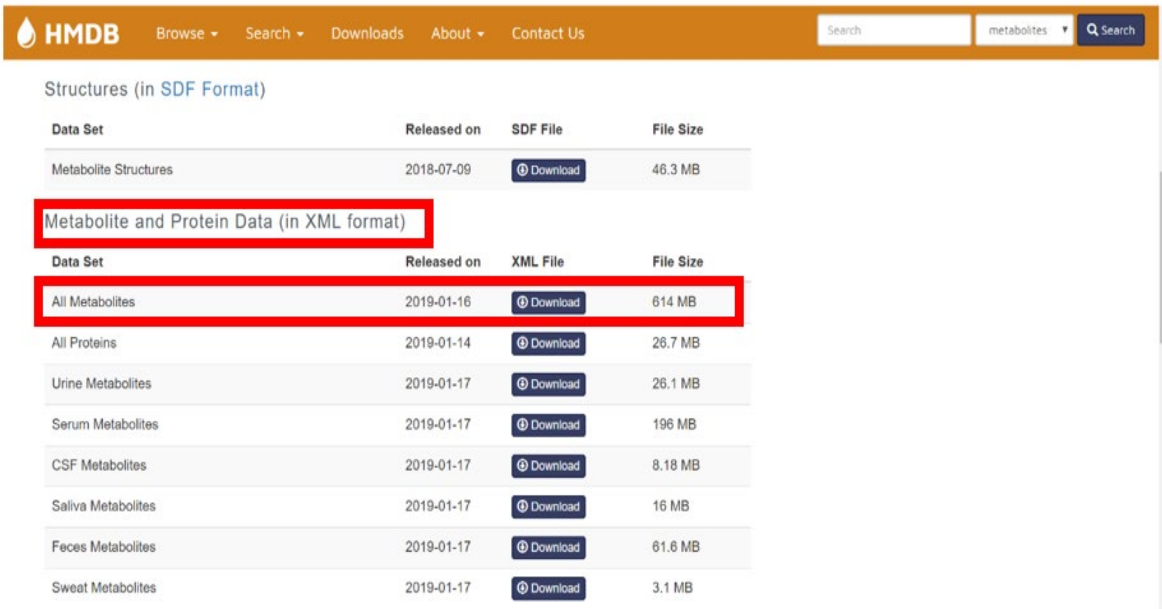
Noted that the datasets excerpted from PubChem and HMDB were downloaded in 2018 and thus the contents in the current web pages now may be different from the snapshots below. Also, the file names and the corresponding path may need to be changed in the provided programs. In addition, the provided Python programs are better run under python 3.6 or later and the MATLAB programs are better run under MATLAB 2021b or later.

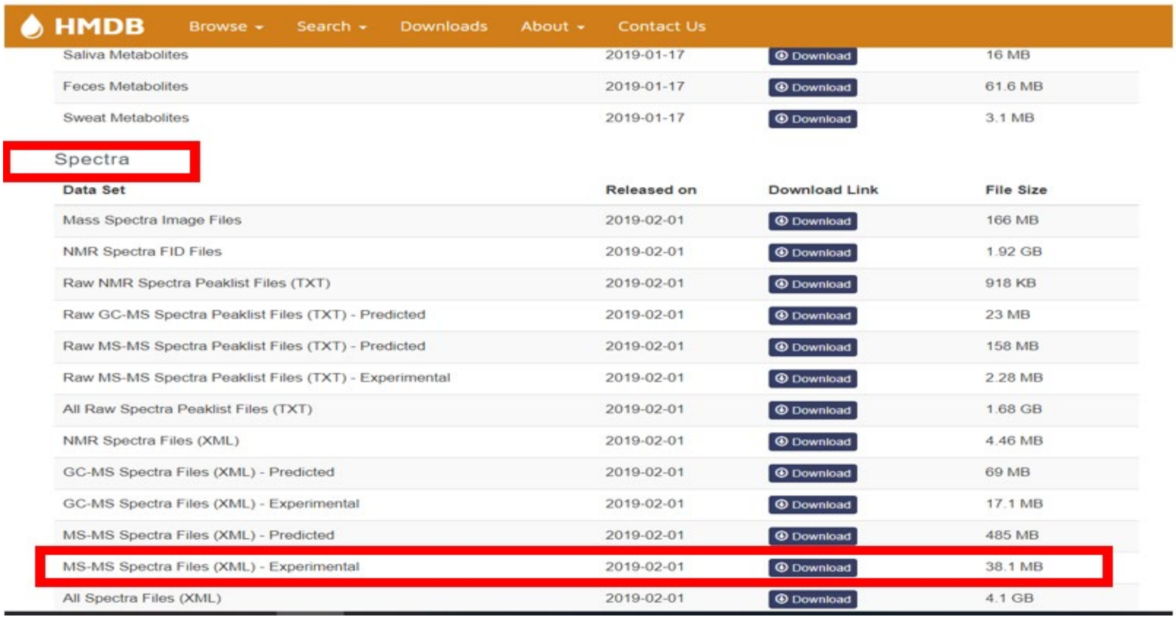
1. Download Data:
   a、 PubChem: For the search of compound mass formula candidates (https://ftp.ncbi.nlm.nih.gov/pubchem/Compound/CURRENT-Full/XML/)



   b、 HMDB Test dataset (http://www.hmdb.ca/downloads)
      i. Compound information



      ii. MS/MS spectra



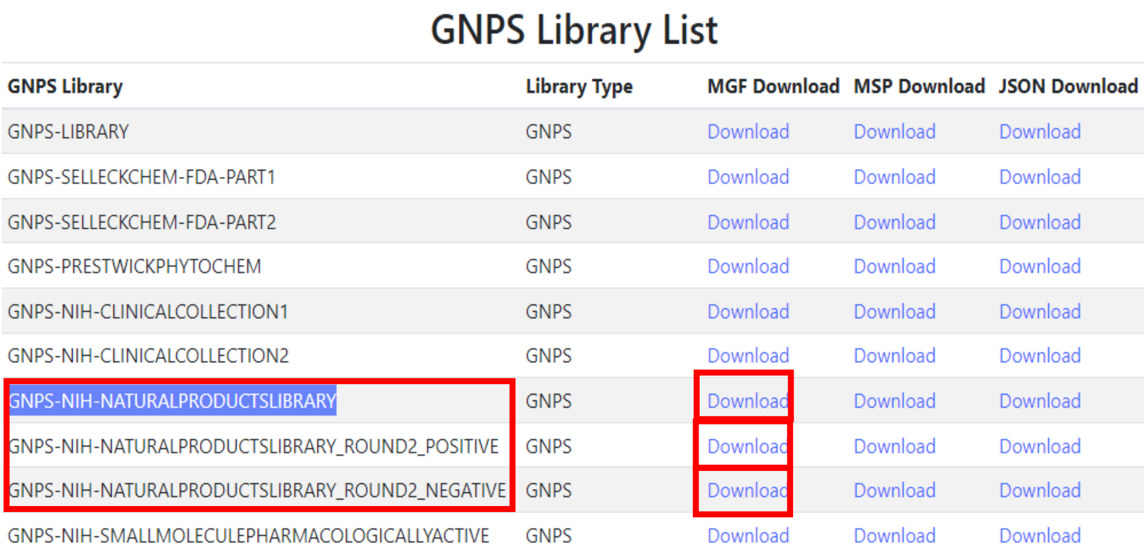
Saliva Metabolites	2019-01-17	Download	16 MB
Feces Metabolites	2019-01-17	Download	61.6 MB
Sweat Metabolites	2019-01-17	Download	3.1 MB

Spectra

Data Set	Released on	Download Link	File Size
Mass Spectra Image Files	2019-02-01	Download	166 MB
NMR Spectra FID Files	2019-02-01	Download	1.92 GB
Raw NMR Spectra Peaklist Files (TXT)	2019-02-01	Download	918 KB
Raw GC-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	23 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	158 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Experimental	2019-02-01	Download	2.28 MB
All Raw Spectra Peaklist Files (TXT)	2019-02-01	Download	1.68 GB
NMR Spectra Files (XML)	2019-02-01	Download	4.46 MB
GC-MS Spectra Files (XML) - Predicted	2019-02-01	Download	69 MB
GC-MS Spectra Files (XML) - Experimental	2019-02-01	Download	17.1 MB
MS-MS Spectra Files (XML) - Predicted	2019-02-01	Download	485 MB
MS-MS Spectra Files (XML) - Experimental	2019-02-01	Download	38.1 MB
All Spectra Files (XML)	2019-02-01	Download	4.1 GB

c、 GNPS NIH Natural Products Library:

Go to the website at https://gnps-external.ucsd.edu/gnpslibrary, find the rows shown in the following figure, and click on the "Download" links under the "MGF Download" column.

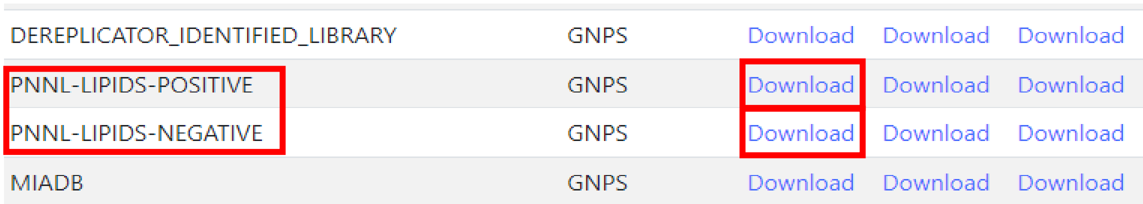


GNPS Library List

GNPS Library	Library Type	MGF Download	MSP Download	JSON Download
GNPS-LIBRARY	GNPS	Download	Download	Download
GNPS-SELLECKCHEM-FDA-PART1	GNPS	Download	Download	Download
GNPS-SELLECKCHEM-FDA-PART2	GNPS	Download	Download	Download
GNPS-PRESTWICKPHYTOCHEM	GNPS	Download	Download	Download
GNPS-NIH-CLINICALCOLLECTION1	GNPS	Download	Download	Download
GNPS-NIH-CLINICALCOLLECTION2	GNPS	Download	Download	Download
GNPS-NIH-NATURALPRODUCTSLIBRARY	GNPS	Download	Download	Download
GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_POSITIVE	GNPS	Download	Download	Download
GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_NEGATIVE	GNPS	Download	Download	Download
GNPS-NIH-SMALLMOLECULEPHARMACOLOGICALLYACTIVE	GNPS	Download	Download	Download

d、 GNPS Pacific Northwest National Lab Lipids Library

Go to the website at https://gnps-external.ucsd.edu/gnpslibrary, find the rows shown in the following figure and click on the "Download" links under the "MGF Download" column.



DEREPLICATOR_IDENTIFIED_LIBRARY	GNPS	Download	Download	Download
PNNL-LIPIDS-POSITIVE	GNPS	Download	Download	Download
PNNL-LIPIDS-NEGATIVE	GNPS	Download	Download	Download
MIADB	GNPS	Download	Download	Download

2. Use **MySQL Workbench** to construct a **compound** database
   a、 Install **MySQL Workbench** and create a user "root" with password "1234" (MySQL Workbench Installation: A Step-by-Step Guide https://www.simplilearn.com/tutorials/mysql-tutorial/mysql-workbench-installation).
   b、 Add a "**compound"** database
      i. Right-click on the list of existing **Schemas** and select **Create Schema.**
      ii. Enter "**compound**" for the schema and for collation choose '**utf - utf8_bin**'. Then click **Apply**.
      iii. In the **Apply SQL Script to Database** window, click **Apply** to run the SQL command that creates the schema.
      iv. Click **Finish**. You can see the new schema, which has no tables, listed in the left pane.
      v. From the **Management** menu, select **Users and Privileges** and click **Add Account**. Complete the screen with the credentials listed above. Navigate to the tab **Schema Privileges** and click on **Add Entry...**. Select the newly created database schema. Click **Select "All"** to grant all privileges on this schema for this new user. Click **Apply**.
      vi. From the **Management** menu, select **Options File** and click the **Networking** tab. Find the max_allowed_packet entry (should be at the top) and change it to at least 256M. Click **Apply...** and in the new dialog that appears again **Apply**.
      vii. If you get the error *Could not save configuration file*, you did not run the MySQL Workbench as an administrator. Restart it as an administrator and try again.
      viii. Finally, to apply this change you need to restart the database. From the **Instance** menu, select **Startup/Shutdown** and click **Stop Server**, followed by **Start Server**.
3. Construct PubChem Dataset for Compound Formula Candidates Search
   a、 Use the provided readgzip_COMPOUNDtoMYSQL.py to convert PubChem XML files into MySQL database table "**compounddata20190529**".
   b、 Be sure to adjust the user name and password of the SQL server and the file names and paths of the downloaded files if different names are used.
   c、 Use the following query to extract the qualified records and save them into "PubChem_metabolites.csv"
   SELECT MolecularFormula, Weight_MonoIsotopic
   FROM compounddata20190529 as t1
   WHERE t1.Weight_MonoIsotopic < 1500 and (t1.MolecularFormula like '%C%' or t1.MolecularFormula like '%H%' or t1.MolecularFormula like '%O%' or t1.MolecularFormula like '%N%' or t1.MolecularFormula like '%S%' or t1.MolecularFormula like '%P%') and t1.MolecularFormula not like '%F%' and t1.MolecularFormula not like '%Cl%' and t1.MolecularFormula not like '%Br%' and t1.MolecularFormula not like '%I%'
   INTO OUTFILE '**your_SQL_directory**/Uploads/PubChem_metabolites.csv'

FIELDS TERMINATED BY ','
LINES TERMINATED BY '\r\n';

d、 Use Matlab program "convert_PubChem_metabollites.m" to further ensure the element contains CHONSP only and to convert the formula into a numerical matrix indicating number of CHONSP elements in each formula. The "PubChem_metabolites.csv" is converted into output file "PubChemMetabolite.mat" which is need in all the FSA tests in step 5.

4. Construct test datasets

a、 HMDB

i. Use the provided python file HMDBtoMySQL4_0218.py to convert HMDB Compound XML file into MySQL database table "**hmdbdata20190226**".

ii. Use the provided python files MSMStoMySQL.py and MSMStoMySQL_otherproblem.py to convert HMDB MS/MS XML file into MySQL database table "msmsdata20190226"

iii. Use the follow query to extract the qualified records and save them to " HMDB_CHONPS_MSMS.txt". Note that your_MySQL_upload_directory can be obtained by the query "**SHOW GLOBAL VARIABLES LIKE '%secure%'**"

```
select 'name', 'accession', 'chemical_formula', 'monisotopic_molecular_weight',
'ionization_mode', 'instrument_type', 'collision_energy_voltage', 'peak_counter',
'ms2_peak'
union all
select t1.name, t1.accession, t1.chemical_formula, t1.monisotopic_molecular_weight,
t2.ionization_mode, t2.instrument_type,   t2.collision_energy_voltage, t2.peak_counter,
t2.ms2_peak
from hmdbdata20190226 as t1
inner join msmsdata20190226 as t2
on t2.database_id=t1.accession
where (t2.instrument_type like '%LC-ESI-QTOF%' or t2.instrument_type like '%LC-ESI-
ITFT%' or t2.instrument_type like '%LC-ESI-QFT%') and t2.peak_counter > 2 and
t1.chemical_formula like '%C%' and t1.chemical_formula like '%H%' and
t1.chemical_formula not like '%F%' and t1.chemical_formula not like '%Cl%' and
t1.chemical_formula not like '%Br%' and t1.chemical_formula not like '%I%' and
t1.chemical_formula not like'%B%' and t1.chemical_formula not like'%Si%'
into outfile 'your_MySQL_upload_directory\\HMDB_CHONPS_MSMS.txt'
FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\r\n';
```

iv. Use the provided "divide_HMDB_instruments.m" to divide HMDB_CHONPS_MSMS.txt into 3 Matlab data files HMDB_QTOF_spectra.mat, HMDB_ITFT_spectra.mat, and HMDB_QFT_spectra.mat.

b、 GNPS NIH Natural Products Library:

i. Put the provided 3 Matlab files GNPS_NPL_mgf2mat.m, readNatProdMGF.m, and readGNPS_NatProdR2_MGF.m along with the 3 data files GNPS-NIH-NATURALPRODUCTSLIBRARY.mgf, GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_NEGATIVE.mgf, and GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_POSITIVE.mgf downloaded in step 1(c) into the same directory.

ii. Run GNPS_NPL_mgf2mat.m in Matlab (2021a or a later version) and you should have 3 Matlab data files GNPS_NPL_MAXIS.mat, GNPS_NPL_QTOF.mat, and GNPS_NPL_ORBITRAP.mat in the directory.

c、 GNPS Pacific Northwest National Lab Lipids Library:

i. Put the provided 3 Matlab files GNPS_PNNL_LIPIDS_mgf2mat.m and readGNPS_PNNL_LIPIDS_MGF.m along with the 2 data files PNNL-LIPIDS-NEGATIVE.mgf and PNNL-LIPIDS-POSITIVE.mgf downloaded in step 1(d) into the same directory.

ii. Run GNPS_PNNL_LIPIDS_mgf2mat.m in Matlab (2021a or a later version) and you should have the Matlab data file PNNLLIPID_spectra.mat in the directory.

5. Run FSA on the test datasets in Matlab (2021a or a later version)

a、 Use the provided FSA_all7datasets.m and put the 8 previous generated PubChemMetabolite.mat, HMDB_QTOF_spectra.mat, HMDB_QTOF_spectra.mat, HMDB_QTOF_spectra.mat, GNPS_NPL_MAXIS.mat, GNPS_NPL_QTOF.mat, GNPS_NPL_ORBITRAP.mat, and PNNLLIPID_spectra.mat in the same directory. Users can download the provided FSA_HMDB_result_CHONSP.xlsx and PubChemMetabolite.mat since they are authorized to be distributed here.

b、 Run FSA_all7datasets.m and you should have the output file Supplementary_File1.xlsx, which includes the formula ranking results of the 7 datasets.

6. Run FSA on user's dataset in Matlab (2021a or later version)

a、 A single peak list from a MS/MS spectrum:

i. Modify lines 14~19 of the provided SingleSpectrumFSA.m in Matlab

Line14: The precursor m/z value

Line15: The m/z values of the peaks in a centroided MS/MS spectrum

Line16: The corresponding abundances of the peaks (in percentage) in line 15

Line17: Mass matching tolerance (can be either in Da (if value < 1) or in PPM (if value ≥ 1))

Line18: Max number of ranked formulas for each spectrum

```
14    precursor_mz=245.2341;
15    mz=[245.2341, 171.1502, 129.1395, 112.1128, 100.0760, 84.0812];
16    ab=[56.8106, 58.1811, 65.6561, 90.0748, 100.0000, 51.4535];
17    mode='positive';
18    Match_Tol=0.001;
19    MaxQualifiedResult=5;
```

ii. Run the modified program in Matlab

b、 Multiple peak lists from a MS/MS spectral data set:

i. Generate a MS Excel file of the peak lists. Note that in a peak list, m/z and abundance (or intensity) values should appear in pairs and all values should be separated by a semicolon (";"). An example file "MSMS_peaklist_example.xlsx" is provided.

ii. Modify lines 21~29 of the provided "MultipleSpectraFSA.m".

Line21: Threshold (in ratio) to remove small peaks

Line22: mass matching tolerance

Line23: Max number of ranked formulas for each spectrum

Line27: File name of the peak lists as generated in step i.

Line29: File name of the output formula ranking result.

```
17  % %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
18  % Please fill in the following required information
19  % %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
20  % Parameter settings
21  opt.TINY=0.5; % ratio threshold for "tiny" peaks
22  opt.MatchTolerance=0.001;% mass match tolerance (can be set in the unit of either PPM or Da)
23  opt.MaxRankNumber=3;
24  % %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
25  % Please provide your peak list and output filename
26  % %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
27  MS2Table=readtable('MSMS_peaklist_example.xlsx','FileType','spreadsheet',...
28     'TextType','string','VariableNamingRule','modify'); % file name of the spectral peak list
29  output_filename='Precursor_Formula_Ranking_Result_CHONSP.xlsx'; % file name of the formula ranking result
30  % %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

iii. Run the modified program in Matlab